
```
1  // A program that says hello to the world
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      printf("hello, world\n");
8  }
```

```
1 // get_string and printf with incorrect placeholder
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     string answer = get_string("What's your name? ");
9     printf("hello, answer\n");
10 }
```

```
1 // get_string and printf with %s
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     string answer = get_string("What's your name? ");
9     printf("hello, %s\n", answer);
10 }
```

```
1  // Conditional, Boolean expression, relational operator
2
3  #include <cs50.h>
4  #include <stdio.h>
5
6  int main(void)
7  {
8      // Prompt user for integers
9      int x = get_int("What's x? ");
10     int y = get_int("What's y? ");
11
12     // Compare integers
13     if (x < y)
14     {
15         printf("x is less than y\n");
16     }
17 }
```

```
1 // Conditionals that are mutually exclusive
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for integers
9     int x = get_int("What's x? ");
10    int y = get_int("What's y? ");
11
12    // Compare integers
13    if (x < y)
14    {
15        printf("x is less than y\n");
16    }
17    else
18    {
19        printf("x is not less than y\n");
20    }
21 }
```

```
1 // Conditionals that aren't mutually exclusive
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for integers
9     int x = get_int("What's x? ");
10    int y = get_int("What's y? ");
11
12    // Compare integers
13    if (x < y)
14    {
15        printf("x is less than y\n");
16    }
17    if (x > y)
18    {
19        printf("x is greater than y\n");
20    }
21    if (x == y)
22    {
23        printf("x is equal to y\n");
24    }
25 }
```

```
1 // Conditional that isn't necessary
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for integers
9     int x = get_int("What's x? ");
10    int y = get_int("What's y? ");
11
12    // Compare integers
13    if (x < y)
14    {
15        printf("x is less than y\n");
16    }
17    else if (x > y)
18    {
19        printf("x is greater than y\n");
20    }
21    else if (x == y)
22    {
23        printf("x is equal to y\n");
24    }
25 }
```

```
1 // Conditionals
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for integers
9     int x = get_int("What's x? ");
10    int y = get_int("What's y? ");
11
12    // Compare integers
13    if (x < y)
14    {
15        printf("x is less than y\n");
16    }
17    else if (x > y)
18    {
19        printf("x is greater than y\n");
20    }
21    else
22    {
23        printf("x is equal to y\n");
24    }
25 }
```



```
1  // Comparing against lowercase char
2
3  #include <cs50.h>
4  #include <stdio.h>
5
6  int main(void)
7  {
8      // Prompt user to agree
9      char c = get_char("Do you agree? ");
10
11     // Check whether agreed
12     if (c == 'y')
13     {
14         printf("Agreed.\n");
15     }
16     else if (c == 'n')
17     {
18         printf("Not agreed.\n");
19     }
20 }
```

```
1 // Comparing against lowercase and uppercase char
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user to agree
9     char c = get_char("Do you agree? ");
10
11     // Check whether agreed
12     if (c == 'y')
13     {
14         printf("Agreed.\n");
15     }
16     else if (c == 'Y')
17     {
18         printf("Agreed.\n");
19     }
20     else
21     {
22         printf("Not agreed.\n");
23     }
24 }
```

```
1 // Logical operators
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user to agree
9     char c = get_char("Do you agree? ");
10
11     // Check whether agreed
12     if (c == 'Y' || c == 'y')
13     {
14         printf("Agreed.\n");
15     }
16     else
17     {
18         printf("Not agreed.\n");
19     }
20 }
```

```
1  // Opportunity for better design
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      printf("meow\n");
8      printf("meow\n");
9      printf("meow\n");
10 }
```

```
1 // Better design
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     int i = 3;
8     while (i > 0)
9     {
10         printf("meow\n");
11         i--;
12     }
13 }
```

```
1 // Print values of i
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     int i = 1;
8     while (i <= 3)
9     {
10         printf("meow\n");
11         i++;
12     }
13 }
```

```
1  // Better design
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      int i = 0;
8      while (i < 3)
9      {
10         printf("meow\n");
11         i++;
12     }
13 }
```

```
1  // Better design
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      for (int i = 0; i < 3; i++)
8      {
9          printf("meow\n");
10     }
11 }
```

```
1 // Infinite loop
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     while (true)
9     {
10         printf("meow\n");
11     }
12 }
```

```
1 // Prompts user for n.
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     int n = get_int("What's n? ");
9
10    for (int i = 0; i < n; i++)
11    {
12        printf("meow\n");
13    }
14 }
```

```
1 // Prompts user again if need be. (Poor design.)
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     int n = get_int("What's n? ");
9     if (n < 0)
10     {
11         n = get_int("What's n? ");
12     }
13
14     for (int i = 0; i < n; i++)
15     {
16         printf("meow\n");
17     }
18 }
```

```
1 // Uses a loop with continue/break.
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     int n;
9     while (true)
10     {
11         n = get_int("What's n? ");
12         if (n < 0)
13         {
14             continue;
15         }
16         else
17         {
18             break;
19         }
20     }
21
22     for (int i = 0; i < n; i++)
23     {
24         printf("meow\n");
25     }
26 }
```

```
1 // Uses a loop with just break.
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     int n;
9     while (true)
10     {
11         n = get_int("What's n? ");
12         if (n >= 0)
13         {
14             break;
15         }
16     }
17
18     for (int i = 0; i < n; i++)
19     {
20         printf("meow\n");
21     }
22 }
```

```
1  // Uses a do-while loop instead.
2
3  #include <cs50.h>
4  #include <stdio.h>
5
6  int main(void)
7  {
8      int n;
9      do
10     {
11         n = get_int("What's n? ");
12     }
13     while (n < 0);
14
15     for (int i = 0; i < n; i++)
16     {
17         printf("meow\n");
18     }
19 }
```

```
1  // Abstraction
2
3  #include <stdio.h>
4
5  void meow(void);
6
7  int main(void)
8  {
9      for (int i = 0; i < 3; i++)
10     {
11         meow();
12     }
13 }
14
15 // Meow once
16 void meow(void)
17 {
18     printf("meow\n");
19 }
```

```
1  // Abstraction with parameterization
2
3  #include <stdio.h>
4
5  void meow(int n);
6
7  int main(void)
8  {
9      meow(3);
10 }
11
12 // Meow some number of times
13 void meow(int n)
14 {
15     for (int i = 0; i < n; i++)
16     {
17         printf("meow\n");
18     }
19 }
```



```
1  // Demonstrates scope
2
3  #include <stdio.h>
4
5  void meow(int n);
6
7  int main(void)
8  {
9      int n = 3;
10     meow(n);
11 }
12
13 // Meow some number of times
14 void meow(int n)
15 {
16     for (int i = 0; i < n; i++)
17     {
18         printf("meow\n");
19     }
20 }
```

```
1 // User input
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 void meow(int n);
7
8 int main(void)
9 {
10     int n;
11     do
12     {
13         n = get_int("Number: ");
14     }
15     while (n < 1);
16     meow(n);
17 }
18
19 // Meow some number of times
20 void meow(int n)
21 {
22     for (int i = 0; i < n; i++)
23     {
24         printf("meow\n");
25     }
26 }
```

```
1  // Return value
2
3  #include <cs50.h>
4  #include <stdio.h>
5
6  int get_positive_int(void);
7  void meow(int n);
8
9  int main(void)
10 {
11     int n = get_positive_int();
12     meow(n);
13 }
14
15 // Get number of meows
16 int get_positive_int(void)
17 {
18     int n;
19     do
20     {
21         n = get_int("Number: ");
22     }
23     while (n < 1);
24     return n;
25 }
26
27 // Meow some number of times
28 void meow(int n)
29 {
30     for (int i = 0; i < n; i++)
31     {
32         printf("meow\n");
33     }
34 }
```

```
1 // Prints a row of 4 question marks
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     printf("????\n");
8 }
```

```
1  // Prints a row of 4 question marks with a loop
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      for (int i = 0; i < 4; i++)
8      {
9          printf("?");
10     }
11     printf("\n");
12 }
```

```
1  // Prints a column of 3 bricks with a loop
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      for (int i = 0; i < 3; i++)
8      {
9          printf("#\n");
10     }
11 }
```

```
1  // Prints a 3-by-3 grid of bricks with nested loops
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      for (int i = 0; i < 3; i++)
8      {
9          for (int j = 0; j < 3; j++)
10         {
11             printf("#");
12         }
13         printf("\n");
14     }
15 }
```

```
1  // Prints a 3-by-3 grid of bricks with nested loops using a constant
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      const int n = 3;
8      for (int i = 0; i < n; i++)
9      {
10         for (int j = 0; j < n; j++)
11         {
12             printf("#");
13         }
14         printf("\n");
15     }
16 }
```



```
1 // Helper function
2
3 #include <stdio.h>
4
5 void print_row(int width);
6
7 int main(void)
8 {
9     const int n = 3;
10    for (int i = 0; i < n; i++)
11    {
12        print_row(n);
13    }
14 }
15
16 void print_row(int width)
17 {
18     for (int i = 0; i < width; i++)
19     {
20         printf("#");
21     }
22     printf("\n");
23 }
```

```
1 // Addition with int
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for x
9     int x = get_int("What's x? ");
10
11     // Prompt user for y
12     int y = get_int("What's y? ");
13
14     // Add numbers
15     int z = x + y;
16
17     // Perform addition
18     printf("%i\n", z);
19 }
```

```
1 // Addition with int, without third variable
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for x
9     int x = get_int("What's x? ");
10
11     // Prompt user for y
12     int y = get_int("What's y? ");
13
14     // Perform addition
15     printf("%i\n", x + y);
16 }
```

```
1 // Doubles a number
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for x
9     int x = get_int("What's x? ");
10
11     // Double it
12     printf("%i\n", x * 2);
13 }
```

```
1  // Overflow
2
3  #include <cs50.h>
4  #include <stdio.h>
5
6  int main(void)
7  {
8      int dollars = 1;
9      while (true)
10     {
11         char c = get_char("Here's $%i. Double it and give to next person? ", dollars);
12         if (c == 'y')
13         {
14             dollars *= 2;
15         }
16         else
17         {
18             break;
19         }
20     }
21     printf("Here's $%i.\n", dollars);
22 }
```

```
1 // long
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     long dollars = 1;
9     while (true)
10     {
11         char c = get_char("Here's $%li. Double it and give to next person? ", dollars);
12         if (c == 'y')
13         {
14             dollars *= 2;
15         }
16         else
17         {
18             break;
19         }
20     }
21     printf("Here's $%li.\n", dollars);
22 }
```

```
1 // Division with ints, demonstrating truncation
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for x
9     int x = get_int("What's x? ");
10
11     // Prompt user for y
12     int y = get_int("What's y? ");
13
14     // Divide x by y
15     printf("%i\n", x / y);
16 }
```

```
1 // Casting
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for x
9     int x = get_int("What's x? ");
10
11     // Prompt user for y
12     int y = get_int("What's y? ");
13
14     // Divide x by y
15     printf("%f\n", (float) x / y);
16 }
```



```
1 // Floats
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Prompt user for x
9     float x = get_float("What's x? ");
10
11     // Prompt user for y
12     float y = get_float("What's y? ");
13
14     // Divide x by y
15     printf("%.50f\n", x / y);
16 }
```